Chapter 21: Structural Equation Modeling

1. Introduction

Structural Equation Modeling (SEM) examines causal relationships through path analysis. Effects decompose into direct (unmediated) and indirect (mediated) pathways. SEM integrates measurement and structural models.

2. Theoretical Model

Self-efficacy influences motivation and course engagement, which jointly affect course performance. Self-efficacy also has a direct effect on performance, net of motivation and engagement.

3. Methods

Data: 400 respondents. Regression analysis estimated path coefficients. Total, direct, and indirect effects calculated.

4. Results

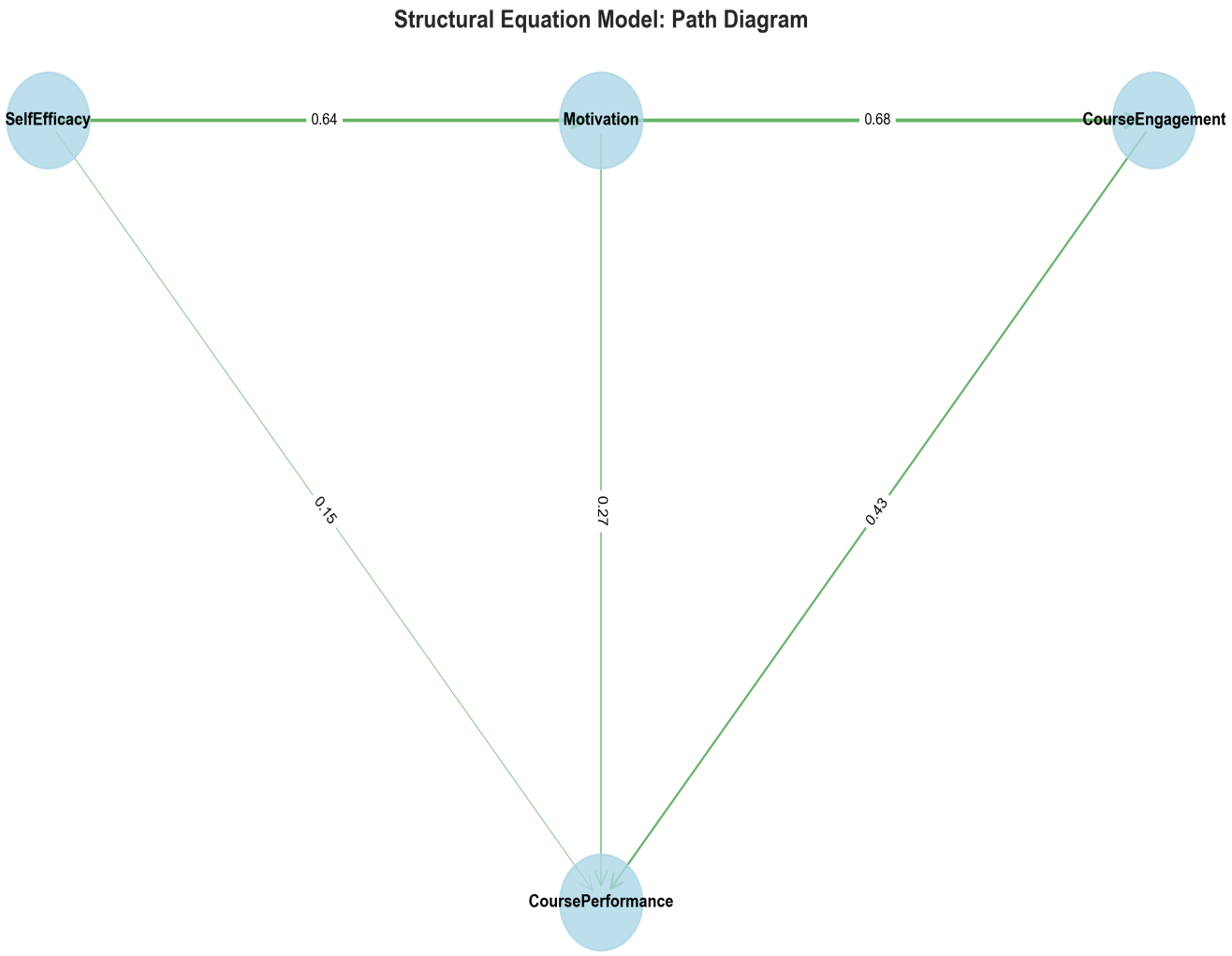

Fig 1: Path diagram with standardized coefficients.

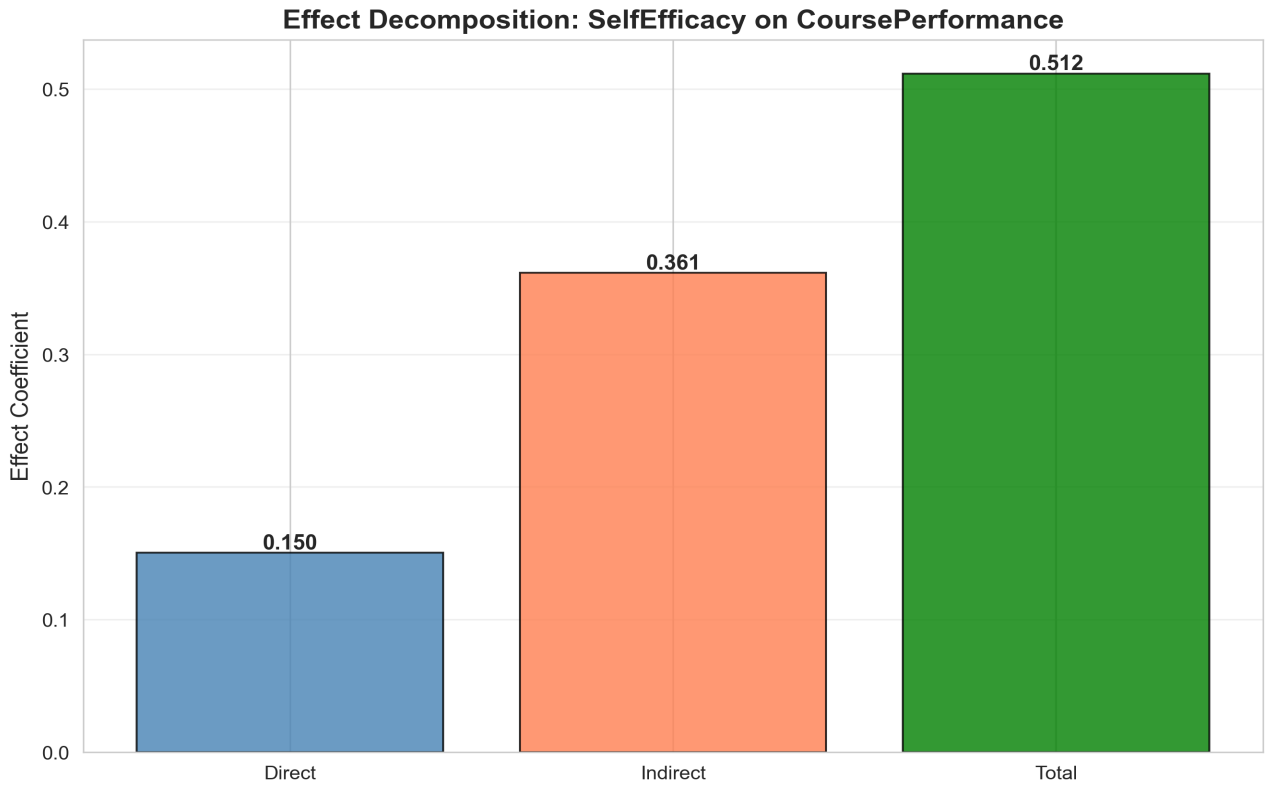

Fig 2: Effect decomposition showing mediation.

5. Key Findings

Direct effect of Self-Efficacy on Performance: 0.150. Indirect effect through Motivation and Engagement: 0.361. Total effect: 0.512. Results suggest significant mediation pathways.

6. Limitations

Path analysis assumes linear relationships and single indicators per construct. Full SEM would include latent variable measurement models.

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